

उम्मीदवार इस पुस्तिका के सबसे ऊपरी सील को खोलकर पृष्ठ संख्या 2 और 3 के मध्य स्थापित OMR उत्तर पुस्तिका को निकाल लें।
Candidates should open the top side of the seal of this Booklet and take out the OMR Answer Sheet placed between page no. 2 and 3.

पुस्तिका सं.:
Test Booklet No.:

11949

परीक्षा पुस्तिका श्रृंखला : A
Test Booklet Series :

निर्धारित समय : 150 मिनट
Time Allowed : 150 Minutes

परीक्षा प्रश्न-पुस्तिका EXAMINATION QUESTION BOOKLET

अधिकतम अंक : 150
Maximum Marks : 150

रोल नं.:
Roll No.:

C1

उत्तर शीट सं.:
Answer Sheet No.:

प्रश्नों के उत्तर देने से पहले निम्नलिखित अनुदेशों को ध्यान से पढ़ लें।
Read the following instructions carefully before you begin to answer the questions.

उम्मीदवारों के लिए अनुदेश

Instructions to the Candidates

- प्रश्नों के उत्तर लिखना आरंभ करने से पहले आप इस पुस्तिका की जाँच करके सुनिश्चित कर लें कि इसमें पूरे पृष्ठ (1-24) हैं तथा कोई पृष्ठ या उसका भाग कम या दुबारा तो नहीं आ गया है। उम्मीदवारों को यह भी जाँच करनी है कि उनको केवल उस पोस्ट की सही परीक्षा-पुस्तिका मिली है जिसके लिए उन्होंने आवेदन किया है। यदि आप इस पुस्तिका में कोई त्रुटि पाएँ, तो तत्काल इसके बदले दूसरी पुस्तिका लें।
- ओ एम् आर उत्तर-शीट प्रश्न पुस्तिका में ही उपलब्ध रहेगी। कृपया सुनिश्चित करें कि ओ एम् आर उत्तर-शीट संख्या और परीक्षा पुस्तिका संख्या समान हैं। ओ एम् आर उत्तर-शीट पर जानकारी भरने से पहले ओ एम् आर शीट पर छपे निर्देशों को ध्यान से पढ़ें। आपको ओ एम् आर उत्तर पत्रक पर सभी विवरणों को सही ढंग से पूरा और कोड करना होगा, ऐसा न करने पर आपकी उत्तर पुस्तिका का मूल्यांकन नहीं किया जा सकता है। प्रश्नों का उत्तर देना शुरू करने से पहले आपको ओ एम् आर उत्तर-पत्रक पर दिये गए निर्धारित स्थान पर अपने हस्ताक्षर करने होंगे। इन निर्देशों का पूर्ण रूप से पालन किया जाना चाहिए, ऐसा न करने पर आपकी ओ एम् आर उत्तर-पुस्तिका का मूल्यांकन नहीं किया जा सकता है। (दृष्टिहीन उम्मीदवारों के लिए यह विवरण लेखक द्वारा भरे जायेंगे। फिर भी, सभी दृष्टिहीन उम्मीदवारों को ओ एम् आर उत्तर-शीट में निर्धारित स्थान पर अपने बाएँ हाथ के अंगूठे का निशान अवश्य लगाना चाहिए। इसके अतिरिक्त, जो दृष्टिहीन उम्मीदवार अपना हस्ताक्षर कर सकते हैं, वे अंगूठे के निशान के अलावा अपने हस्ताक्षर भी करें।)
- ओ एम् आर उत्तर-शीट तीन प्रतियों में होगी (मूल तथा कार्बन की दो प्रतिलिपियाँ)। परीक्षा समाप्ति के बाद ओ.एम्.आर. की मूल शीट तथा एक कार्बन प्रतिलिपि निरीक्षक को सौंपने के पश्चात् उम्मीदवार अपने साथ एक कार्बन प्रतिलिपि ले जा सकते/सकती हैं। यदि कोई भी उम्मीदवार ऐसा करने में असफल रहता/रहती है तो उसकी उम्मीदवारी रद्द कर दी जायेगी। यदि कोई उम्मीदवार अपनी कार्बन प्रतिलिपि में किसी भी प्रकार का फेर-बदल कर उसका दावा करता/करती है तो इस स्थिति में भी उसका/उसकी उम्मीदवारी रद्द की जायेगी।
- इस प्रश्न-पुस्तिका में 150 बहुविकल्पीय प्रश्न हैं, प्रत्येक प्रश्न के 4 विकल्प दिए गए हैं, (A), (B), (C) और (D)। किसी भी स्थिति में प्रत्येक प्रश्न का केवल एक विकल्प ही सही उत्तर है। यदि आपको एक से अधिक विकल्प सही लगें तो सबसे अधिक उचित एक विकल्प का चुनाव करें और उत्तर शीट में संबंधित प्रश्न के सामने वाले उपयुक्त गोले को काला करें।
- प्रश्न पुस्तिका में पाँच भाग हैं : भाग A : Reasoning, भाग B : English, भाग C : Aptitude, भाग D : Basic Mathematics, भाग E : Attitude & Leadership प्रत्येक भाग में 30 प्रश्न हैं। उम्मीदवार को सभी भागों के उत्तर लिखना अनिवार्य हैं।
- प्रत्येक सही उत्तर के लिए 1 अंक दिया जाएगा। गलत उत्तर के लिए कोई नकारात्मक अंकन नहीं है।
- गोले को काला करने के लिए केवल काले/नीले बॉल प्वाइंट पेन का प्रयोग करें। गोले को एक बार काला करने के बाद इसको मिटाने या बदलने की अनुमति नहीं है। यदि किसी प्रश्न के सामने एक से ज्यादा गोले काले किये गए हों तो मशीन द्वारा उसके लिए शून्य अंक दिया जाएगा।
- किसी भी स्थिति में उत्तर शीट को न मोड़ें।
- उत्तर-पुस्तिका पर कोई भी रफ कार्य नहीं करना है। रफ कार्य के लिए इस पुस्तिका में स्थान दिया गया है।
- परीक्षा हॉल/कमरों में मोबाइल फ़ोन तथा बेतार संचार साधन पूरी तरह निषिद्ध हैं। उम्मीदवारों को उनके अपने हित में सलाह दी जाती है कि मोबाइल फ़ोन/किसी अन्य बेतार संचार साधन को स्विच ऑफ़ करके भी अपने पास न रखें। इस प्रावधान का अनुपालन न करने को परीक्षा में अनुचित उपायों का प्रयोग माना जायेगा और उनके विरुद्ध कार्यवाही की जाएगी, जिसमें उनकी उम्मीदवारी रद्द करना भी शामिल है।
- अभ्यर्थी अपनी उत्तर पुस्तिका पर्यवेक्षक को सौंपे बिना और अपने रोल नंबर के सामने उचित स्थान पर उपस्थिति पत्रक पर हस्ताक्षर किए बिना परीक्षा हॉल/कक्ष से बाहर नहीं जा सकता। इसके अलावा अभ्यर्थी को उपस्थिति पत्रक पर हस्ताक्षर करने से पहले यह भी सुनिश्चित करना चाहिए कि बुकलेट नंबर, बुकलेट सीरीज और ओ एम् आर उत्तर पुस्तिका संख्या सही ढंग से लिखी गई हो। ऐसा ना करने पर, ओ एम् आर उत्तर पुस्तिका को अमान्य माना जाएगा/मूल्यांकन नहीं किया जा सकता है।

- Before you start to answer the questions you must check this booklet and ensure that it contains all the pages (1-24) and see that no page or portion thereof is missing or repeated. Candidates are also required to check that they have got the right question booklet strictly for the post candidate has applied for. If you find any defect in this Booklet, you must get it replaced immediately.
- OMR Answer-Sheet is within the Question booklet. Please ensure OMR Answer-Sheet number and Test Booklet No. of Question Paper are same. Read the instructions printed on OMR Answer sheet carefully before filling the information on the OMR Answer sheet. You must complete and code all the details on the OMR answer sheet correctly failing which your answer sheet may not be evaluated. You must also put your signature on the OMR Answer-Sheet at the prescribed place before you actually start answering the questions. These instructions must be fully complied with, failing which, your OMR Answer-Sheet may not be evaluated. (For V.H. candidates these details will be filled in by the scribe. However, all V.H. candidates must put their left-hand thumb impression at the space provided in the OMR Answer-Sheet. In addition, those V.H. candidates who can sign should also put their signatures in addition to thumb impression.)
- The OMR answer sheet will be in triplicate (Original and two carbon copies). Candidate has to take one carbon copy (marked as 'candidate copy') with him/her after examination and handover the original OMR along with one carbon copy to invigilator. If candidate fails to handover the original OMR along with one carbon copy to invigilator, his/her candidature will be cancelled. Further, if the candidate tampers with candidate OMR carbon copy and claims for same, in that case also his/her candidature will be cancelled.
- This booklet consists of 150 Multiple Choice Questions. Each question has 4 (four) alternatives (A), (B), (C) and (D). In any case only one alternative will be the correct answer. In case if you find more than one correct answer, then choose the most appropriate single option and darken the appropriate circle in the answer sheet in front of the related question.
- Question Booklet consists of five Section : Section A : Reasoning, Section B : English, Section C : Aptitude, Section D : Basic Mathematics, Section E : Attitude & Leadership. Each Section has 30 questions. Candidates has to attempt all the parts compulsorily.
- For each correct answer One mark will be given and no negative marking for incorrect answer.
- Use Black/Blue ball point Pen to darken the circle. Answer once darkened is not allowed to be erased or altered. Against any question if more than one circle is darkened, machine will allot zero mark for that question.
- Do not fold answer sheet in any case.
- No rough work is to be done on the Answer Sheet. Space for rough work has been provided in this booklet.
- Mobile phones and wireless communication devices are completely banned in the examination hall/rooms. Candidates are advised not to keep mobile phones/any other wireless communication devices with them even switching it off, in their own interest. Failing to comply with this provision will be considered as using unfair means in the examination and action will be taken against them including cancellation of their candidature.
- Candidate should not leave the examination hall/room without handing over his/her Answer sheet to the invigilator and without signing on the attendance sheet at proper place against your roll number, further candidate should also ensure that booklet no., booklet series and OMR Answer sheet no. are correctly written on attendance sheet before signing on it, failing in doing so, may lead to disqualification/no evaluation of OMR Answer sheet.

जब तक आपसे कहा न जाए तब तक प्रश्न-पुस्तिका न खोलें / DO NOT OPEN THE QUESTION BOOKLET UNTIL YOU ARE TOLD TO DO SO.

उम्मीदवार का नाम/Name of Candidate :

उम्मीदवार के हस्ताक्षर/Signature of Candidate :

SEAL

SECTION A : REASONING

1. Q, R, S, and T are sitting on a bench. P is sitting next to Q, R is sitting next to S, S is not sitting with T who is on the left end of the bench. R is in the second position from the right. P is to the right of Q and T. P and R are sitting together. In which position P is sitting ?
 (A) Between Q and S (B) Between Q and R
 (C) Between T and S (D) Between R and T
2. Chlorophyll : Plant :: Haemoglobin : ?
 (A) Haemorrhage (B) Blood (C) Oxygen (D) Red
3. Here are some words translated from an artificial language.
 malgauper means peach cobbler
 malgauport means peach juice
 moggagrop means apple jelly
 moggasport means apple sweet
 Which word could mean "apple juice"?
 (A) moggaport (B) malgauper (C) gropport (D) moggagrop
4. Pointing to a photograph, a man said, "I have no brother or sister, but that man's father is my father's son." Whose photograph was it ?
 (A) His own (B) His Son
 (C) His Father (D) His Grandfather
5. In a code language if POSE is coded as OQNPRTDF, then the word TYPE will be coded as :
 (A) SUXZOQFD (B) SUXZQOFD
 (C) SUXZOQDF (D) SUXZQODE
6. Complete the series E-5, G-7, I-9, K-11,... ?
 (A) L-13, N-14 (B) M-13, O-15 (C) L-12, M-14 (D) K-12, M-14
7. **Statement (1) :** Pens cost more than pencils.
Statement (2) : Pens cost less than erasers.
Statement (3) : Erasers cost more than pencils and pens.
 If the first two statements are true, the third statement is :
 (A) True (B) False
 (C) Uncertain (D) Cannot be determined
8. 3265 : 4376 :: 4673 : ?
 (A) 2154 (B) 3562 (C) 5487 (D) 5784

9. Four usual dice are thrown on the ground. The total of numbers on the top faces of these four dice is 13 as the top faces showed 4, 3, 1 and 5 respectively. What is the total of the faces touching the ground ?
 (A) 12 (B) 13
 (C) 15 (D) Cannot be determined
10. A woman Manpreet asks a man Harpreet, "You are the brother of my uncle's daughter. How are you related to me?". What was the man's answer ?
 (A) Cousin (B) Son (C) Brother-in-law (D) Nephew
11. Reluctant : Keen :: Remarkable : ?
 (A) Usual (B) Restrained (C) Striking (D) Evocative
12. Rahul has two mothers. Shalini is the step-daughter of Seema and step-sister of Rahul. How is Seema related to Rahul ?
 (A) Step-mother (B) Brother
 (C) Can't be determined (D) None of these
13. Which of the following will come in place of the question mark (?) in the following sequence :
 16D7, 18G10, 21KI4, 25P19, ?
 (A) 30V20 (B) 30V25 (C) 30V24 (D) 29V25
14. Find the next number in the sequence: 16, 30, 54, 88, 132...
 (A) 186 (B) 188 (C) 190 (D) 206
15. A box contains two dozen of glasses. If it is dropped on a concrete floor then which of the following cannot be the ratio of broken and unbroken glasses.
 (A) 2:1 (B) 5:1 (C) 6:1 (D) 3:1
16. Daksh is taller than Manick but not as tall as Rohan. Somesh is shorter than Daksh but taller than Farhan. Who among them is the shortest ?
 (A) Daksh (B) Manick
 (C) Farhan (D) Cannot be determined
17. A river flows west to east and on the way turns left and goes in a semi-circle round a hillock, and then turns left at right angles. In which direction is the river finally flowing ?
 (A) West (B) East (C) North (D) South

18. A woman said, "His wife is the only daughter of my father", after introducing a man. How is that man related to the woman ?
 (A) Husband (B) Brother
 (C) Father-in-law (D) None of these
19. An application was received by class teacher in the afternoon of a week day. Next day she forwarded it to the Student Coordinator, who was on leave that day. The student Coordinator put up the application to the principal next day in the evening. The Principal studied the application and disposed off the matter on the same day, i.e, Saturday. Which day was the application received by the inward clerk ?
 (A) Monday (B) Wednesday (C) Tuesday (D) Thursday
20. If (I) A is brother of B
 (II) B is brother of C
 (III) C is brother of D
 (IV) D is sister of E
 Then which of the following statements is not necessarily true ?
 (A) D is sister of C
 (B) B and C are brothers
 (C) A, B, C are male and D is a female
 (D) D and E are sisters
21. Which letter should be the Tenth letter to the left of the ninth letter from the right, if the first half of the alphabet of English are reversed ?
 (A) D (B) F (C) E (D) I
22. If in a certain language, MADRAS is coded as NBESBT, how is BOMBAY coded in that code ?
 (A) CPNCBX (B) CPNCBZ (C) CPOCBZ (D) CQOCBZ
23. In a family, there are husband-wife, two sons and two daughters. All the ladies were invited to a dinner. Both sons went out to play. Husband did not return from the office. Who was at home ?
 (A) Only wife was at home (B) Nobody was at home
 (C) Only sons were at home (D) All ladies were at home
24. DNN, FPP, HRR, _____, LVV.
 (A) GRR (B) GSS (C) JTT (D) ITT

25. Look at this series : F2, _____, D8, C16, B32. What should fill the blank ?
 (A) A16 (B) G4 (C) E4 (D) E3
26. Sanjeev walks 10 metres towards the South. Turning to the left, he walks 20 metres and then moves to his right. After moving a distance of 20 metres, he turns to the right and walks 20 metres. Finally, he turns to the right and moves a distance of 10 metres. How far and in which direction is he from the starting point ?
 (A) 10 metres North (B) 20 metres South
 (C) 20 metres North (D) 10 metres South
27. In a certain code language, OMNIPRESENT is written as QJONPTSMDRD. How is CREDIBILITY written in that code ?
 (A) DSFEJDDXSHKH (B) JEFSDCXSHKH
 (C) JEFSDDXSHKH (D) JEFSDDZUJMJ

Directions : (28-29) In each of the following letter series, some of the letters are missing which are given in that order as one of the alternatives below it. Choose the correct alternative.

28. _zy_zxy_yxzx_zyx_xy
 (A) yxzyz (B) zxyzy (C) yzxyx (D) xyzyz
29. pqr__rs_rs__s_q_
 (A) spqpprr (B) pqrpppq (C) sqpppqr (D) sqpprrqr
30. According to Wikipedia, 'monologue' is presented by a single character most often to express their mental thoughts aloud, though sometimes also to directly address another character or the audience. Which of the following does **NOT** represent a monologue ?
 (A) A poet reciting a poem on the stage
 (B) Anshuman hosting an award show
 (C) A mother narrating a story to her child
 (D) A teacher taking up student queries in the class

SECTION B : ENGLISH

Direction : (31-34) In the following questions find the appropriate explanation for idioms/phrases mentioned in bold letters.

31. I feel that I have been **left up the creek without a paddle** by my father.

- (A) Left with a large inheritance
- (B) Left without an inheritance
- (C) Left in a pleasant situation
- (D) Left in an unpleasant situation

32. They were beginning to **scrape the barrels**.

- (A) Forced to use one's last resources
- (B) Break into a sudden fight after getting drunk
- (C) To waste one's resources
- (D) Drink without worrying about the consequences

33. **I am beat** generally by the time I reach home.

- (A) Late
- (B) Drunk
- (C) Upbeat and energetic
- (D) Thoroughly fatigued

34. Don't try to **boil the ocean**.

- (A) Attack someone who is immensely powerful
- (B) Try to swindle the high and the mighty
- (C) Try to do something difficult
- (D) Pollute the ocean leading to a rise in its temperature

Direction : (35-37) Choose the correct spelling of the word from the given options.

35. (A) minascule (B) minuscule
(C) miniiscule (D) minnuscule

36. (A) occasionaly (B) occasionally
(C) ocassionally (D) ocassionally

37. (A) flabbergested (B) flabberghasted
(C) flabbergasted (D) flaberghasted

Direction (38-40) : Choose the correct substitute for the following phrases/sentences.

38. One who compiles a dictionary
(A) bibliophile (B) cartographer (C) lexicographer (D) linguist
39. A person who has no money to pay off his debts :
(A) penurious (B) insolvent (C) destitute (D) pauper
40. A political leader who seeks support by appealing to popular desires and prejudices rather than by using rational argument :
(A) patriarch (B) demagogue (C) orator (D) pedagogue

Direction (41-43) : Out of the four sentences given below three are incorrect and one correct. The correct one is your answer.

41. (A) I am working really hard, am I ?
(B) I am working really hard ?
(C) I am working really hard, isn't it ?
(D) Am I working hard enough ?
42. (A) He, you and I will leave for this secret mission tonight.
(B) You, he and I will leave for this secret mission tonight.
(C) I, you and he will leave for this secret mission tonight.
(D) You, I and he will leave for this secret mission tonight.
43. (A) The only people interested in the book seems to be lawyers.
(B) The only people interested in this book are seeming to be lawyers.
(C) The only people interested in the book are seemingly to be lawyers.
(D) The only people interested in the books seem to be lawyers.

Direction (44-46) : Fill in the blanks with the correct option given below.

44. This book comprises _____ five chapters.
(A) no preposition required (B) of
(C) with (D) in
45. Everybody sat _____ the floor, but Mr. Sharma slept _____ a cot _____ the corner.
(A) over-in-on (B) in-over-from
(C) on-on-in (D) about-on-at
46. _____ such brilliant batting.
(A) Never I have seen (B) Rarely I have
(C) Never have I seen (D) Never I had seen

Direction (47-49) : Read the following comprehension carefully and answer the following questions.

Humans are to be blamed for declining bat populations. Bats get poisoned when they consume insects that have been sprayed with pesticides and insecticides. But the biggest problem for bats is the loss of natural habitat. Many bats prefer to roost in dead or dying trees under the loose and peeling bark, or in tree cavities. Some prefer to roost in caverns. Populations have dwindled and diversity has suffered without the protection of these important natural roosts.

47. Find a word in the passage which means the same as **decline**.

- (A) habitat (B) suffered (C) dwindled (D) flourish

48. Which is an appropriate and accurate explication of this paragraph ?

- (A) Bats would thrive if humans did not use synthetic pesticides.
(B) Responsible citizens should construct friendly habitats for wildlife.
(C) Humans have unintentionally contributed to declining bat populations.
(D) Bats must seek roosts in caves and caverns when other roosts are unavailable.

49. Find a word in the paragraph which means the opposite of **coating**.

- (A) peeling (B) protection (C) sprayed (D) cavern

Direction (50-52) : In the following questions choose the word that is similar in meaning to the given word.

50. Tireless

- (A) frugal (B) useless
(C) munificent (D) hard-working

51. Accentuate

- (A) accelerate (B) underscore (C) mask (D) expedite

52. Fastidious

- (A) meticulous (B) vehement (C) ludicrous (D) generous

Direction (53-55) : Choose the word that is opposite in meaning to the given word.

53. Abnegation

- (A) relinquishment (B) eschewal (C) indulgence (D) abstinence

54. Strenuous

- (A) herculean (B) toilsome (C) enervated (D) elusive

55. Object

- (A) remonstrate (B) insult (C) cajole (D) approve

Direction (56 to 60) : Complete the sentences using most appropriate options given below :

56. The orator's fiery speech was the _____ for the ensuing _____.

- (A) Catalyst; revolution (B) Damper; explosion
(C) Epilogue; doctrine (D) Barrier; movement

57. The millionaire added a _____ to his will just before dying.

- (A) Prologue (B) Codicil (C) Formal (D) License

58. The vicious rumours were untrue but still _____ the politician's _____.

- (A) Elevated; status (B) Impugned; reputation
(C) Interrogated; subordinates (D) Alleviated; difficulties

59. Restlessness among the students of colleges and universities has _____ from their deep rooted feeling that their views and aspirations are _____ by their elders.

- (A) arisen, humiliated (B) followed, disregarded
(C) stemmed, ignored (D) started, neglected

60. The _____ of the mountains stood out as craggy _____ on the horizon.

- (A) Colors; shapes (B) Shapes; textures
(C) Outline; shapes (D) Textures; lines

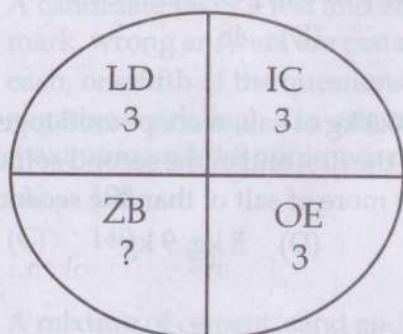
SECTION C : APTITUDE

61. What is the greatest number of 4 digits that when divided by any of the numbers 6, 9, 12, 17 leaves a remainder of 1 ?
 (A) 9997 (B) 9793 (C) 9895 (D) 9487
62. A candidate takes a test and attempts all the 100 questions in it. While any correct answer fetches 1 mark, wrong answers are penalized as follows; one-tenth of the questions carry $1/10$ negative mark each, one-fifth of the questions carry $1/5$ negative marks each and the rest of the questions carry $1/2$ negative mark each. Un-attempted questions carry no marks. What is the difference between the maximum and the minimum marks that he can score ?
 (A) 100 (B) 120
 (C) 140 (D) None of these
63. A mixture of cement, sand and gravel in the ratio of 1:2:4 by volume is required. A person wishes to measure out quantities by weight. He finds that the weight of one cubic foot of cement is 94 kg, of sand 100 kg and gravel 110 kg. What should be the ratio of cement, sand and gravel by weight in order to give a proper mixture ?
 (A) 47:100:220 (B) 94:100:220 (C) 47:200:440 (D) None of these
64. Find the pairs of natural numbers whose least common multiple is 78 and the greatest common divisor is 13.
 (A) 58 and 13 or 16 and 29 (B) 68 and 23 or 36 and 49
 (C) 18 and 73 or 56 and 93 (D) 78 and 13 or 26 and 39
65. A mother divided an amount of ₹ 61,000 between her two daughters aged 18 years and 16 years respectively and deposited their shares in a bond. If the interest rate is 20% compounded annually and if each received the same amount as the other when she attained the age of 20 years, their shares are :
 (A) ₹ 35,600 and ₹ 25,400 (B) ₹ 30500 each
 (C) ₹ 24,000 and ₹ 3700 (D) None of these
66. The sum of two numbers is 20 and their geometric mean is 20% lower than their arithmetic mean. Find the ratio of the numbers.
 (A) 4 : 1 (B) 9 : 1 (C) 1 : 1 (D) 17 : 3
67. How would you best judge the fruitfulness of research ?
 (A) The fruitfulness of any research can best be judged according to whether or not the results are significant.
 (B) The fruitfulness of any research could be judged by assessing the impact of the research on the public or other researchers.
 (C) Fruitfulness of the research is probably best judged in terms of the number of new ideas and insights it offers. This is not easily catalogued; rather it is easy to spot when research lacks novel insight and ideas.
 (D) All of these.

68. How many times would 1 be used while writing all the natural numbers from 8 to 127 in the Binary number system ?
(A) 212 (B) 218 (C) 424 (D) 436
69. Arvind spends 75% of his income and saves the rest. His income is increased by 20% and he increased his expenditure by 10%. Then the increase in savings in percentage is _____.
(A) 55 (B) 52 (C) 50 (D) 48
70. Two solutions, the first of which contains 0.8 kg and the second 0.6 kg of salt, were poured together and 10kg of a new salt solution were obtained. Find the weight of the first and of the second solutions in the mixture if the first solution is known to contain 10 percent more of salt of than the second.
(A) 4 kg, 6 kg (B) 3 kg, 7 kg (C) 4 kg, 9 kg (D) 5 kg, 9 kg
71. Which is the largest of the following fraction ?
(A) $\frac{3}{5}$ (B) $\frac{2}{3}$ (C) $\frac{8}{11}$ (D) $\frac{11}{17}$
72. The size of a rhombus is increased by itself, How many times will its perimeter increase ?
(A) Three times (B) Four times
(C) Two times (D) None of the options
73. If the length of the rectangle is increased by 50% and breadth is decreased by 50% what is the change in the area of the rectangle :
(A) Increase by 12.5% (B) Decrease by 12.5%
(C) No change (D) Decrease by 25%
74. The Product of three consecutive natural numbers, the first of which is an even number, is always divisible by.
(A) 12 (B) 24
(C) 6 (D) All of these
75. Which of the following is a characteristic of qualitative research ?
(A) Design flexibility (B) Inductive analysis
(C) Context sensitivity (D) All of the above
76. Qualitative research methods can be thought of as :
(A) Parallel with the physical sciences.
(B) Methods used to search for the nature of reality.
(C) A stark alternative to quantitative research.
(D) A preliminary stage in the research which can contribute to the development of adequate quantification.

77. Find the number of numbers between 200 and 300, both included, which are not divisible by 2, 3, 4 and 5.
- (A) 27 (B) 26 (C) 25 (D) 28

78. In the given below question, which character when placed at the sign of (?) shall complete the pattern ?



- (A) 3 (B) 16 (C) 5 (D) 13
79. Find the sum of all two digit numbers such that the sum of the digits constituting the number is not less than 7; the sum of the squares of the digits is not greater than 30; the number consisting of the same digits written in the reverse order is not larger than half the given number.
- (A) 52 (B) 51 (C) 49 (D) 53
80. The cost of four mangoes, six guavas and sixteen watermelons is ₹ 500, while the cost of seven mangoes, nine guavas and nineteen watermelons is ₹ 620. What is the cost of one mango, one guava and one watermelon ?
- (A) 120 (B) 40
(C) 150 (D) Cannot be determined
81. Two friends were discussing their marks in an examination. While doing so they realized that both the numbers had the same prime factors, although Raveesh got a score which had two more factors than Harish. If their marks are represented by one of the options as given below, which of the following options would correctly represent the number of marks they got ?
- (A) 30, 60 (B) 20, 80 (C) 40, 80 (D) 20, 60
82. Fourth term of an arithmetic progression is 8. What is the sum of the first 7 terms of the arithmetic progression ?
- (A) 7 (B) 64
(C) 56 (D) Cannot be determined

83. Find the two-digit number the quotient of whose division by the product of its digits is equal to $\frac{8}{3}$, and the difference between the required number and the number consisting of the digits written in the reverse order is 18 ?
- (A) 86 (B) 42
(C) 75 (D) None of these
84. In a 100 m race, Shyam runs at 1.66 m/s. If Shyam gives Sujit a start of 4 m and still beats him by 12 seconds, what is Sujit's speed ?
- (A) 1.11 m/s (B) 0.75 m/s (C) 1.33 m/s (D) 1 km/h
85. A precious stone weighing 35 grams worth ₹ 12,500 is accidentally dropped and gets broken into two pieces having weights in the ratio of 2 : 5. If the price varies as the square of the weight then find the loss incurred.
- (A) ₹ 5750 (B) ₹ 6000 (C) ₹ 5102 (D) ₹ 5000
86. Which one of the following is a composite number ?
- (A) 101 (B) 103 (C) 141 (D) 137
87. Ramesh has total of Rs 56000 in four denominations of currency notes namely Rs 100, Rs 200, Rs 500, and Rs 2,000. How many total currency notes he has, if equal number of currency notes in each denomination are held by him.
- (A) 28 (B) 32 (C) 24 (D) 20
88. Find the sum of all two-digit numbers that give a remainder of 3 when they are divided by 7.
- (A) 686 (B) 676 (C) 666 (D) 656
89. Find the least number that when divided by 16, 18 and 20 leaves a remainder 4 in each case, but is completely divisible by 7.
- (A) 364 (B) 2254 (C) 2964 (D) 2884
90. Three vessels having volumes in the ratio of 1 : 2 : 3 are full of a mixture of coke and soda. In the first vessel, ratio of coke and soda is 2 : 3, in second, 3 : 7 and in third, 1 : 4. If the liquid in all three vessels were mixed in a bigger container, what is the resulting ratio of coke and soda ?
- (A) 4 : 11 (B) 5 : 7 (C) 7 : 11 (D) 7 : 5

SECTION D : BASIC MATHEMATICS

91. One fourth of a herd of camels was seen in forest. Twice the square root of the herd had gone to mountains and remaining 15 camels were seen on the bank of a river. Find the total number of camels.
- (A) 24 (B) 30 (C) 36 (D) 42
92. Two trains going on a parallel line in opposite directions take 10 seconds to cross each other. But if they are going in the same direction the longer train crosses the shorter train in 30 seconds. If the length of the longer train decreased by 50%, the time taken to cross the shorter train while going in the same direction decreases by 8 seconds. The time taken by the longer train to cross a tunnel twice its length, if the difference between the length of the trains is 25 m is :
- (A) 30 sec (B) 24 sec (C) 36 sec (D) 40 sec
93. A dishonest dealer sells the goods at $6\frac{1}{4}\%$ loss on cost price but uses $12\frac{1}{4}\%$ less weight. What is his percentage profit or loss ?
- (A) $7\frac{1}{4}\%$ (B) $4\frac{1}{7}\%$ (C) $7\frac{1}{7}\%$ (D) $4\frac{1}{4}\%$
94. Statement (I) : 579876543 is divisible by 999
Statement (II) : 6800900292 is divisible by 999
- (A) Both (I) and (II) are true (B) Both (I) and (II) are false
(C) (I) is true and (II) is false (D) (I) is false and (II) is true
95. Find the roots of equation $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$: $a+b \neq 0$.
- (A) a, b (B) -a, -b (C) 2a, 3b (D) 3a, 2b

96. The value of the expression $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$ is
- (A) 2 (B) $2 \frac{\sin 20^\circ}{\sin 40^\circ}$ (C) 4 (D) $4 \frac{\sin 20^\circ}{\sin 40^\circ}$
97. Find the number of zeroes in the following multiplication :
- $$5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$$
- (A) 6 (B) 7 (C) 8 (D) 5
98. For the positive integers a, b, c, d , find the minimum value of $(-1)^a + (-1)^b + (-1)^c + (-1)^d$, if $a + b + c + d = 1947$.
- (A) -1 (B) -2 (C) -3 (D) -4
99. In the month of July of a certain year, the average daily expenditure of an organization was Rs. 68. For the first 15 days of the month, the average daily expenditure was Rs. 85 and for the last 17 days, Rs. 51. Find the amount spent by the organization on the 15th of the month.
- (A) Rs. 42 (B) Rs. 36 (C) Rs. 34 (D) Rs. 52
100. Mahesh and Umesh purchase a radio each for the same price and both mark up their respective radios by the same amount. Mahesh gives a discount of Rs.20/- followed by another discount of 20% on the reduced price. While Umesh gives a discount of 20% followed by a discount of Rs 20/- . If Mahesh's profit percent is equal to thrice Umesh's loss percent. The profit (in Rs) of Mahesh on his radio is:
- (A) Rs. 10 (B) Rs. 3 (C) Rs. 15 (D) Rs. 7
101. x, y, z are three positive integers such that x is greater than y & y is greater than z then which of the following is always true :
- (A) $y\%$ of z is less than $x\%$ of y .
 (B) $y\%$ of x is less than $z\%$ of y .
 (C) $z\%$ of x is less than $y\%$ of z .
 (D) All of the above.

102. P, Q, R are three typist who working simultaneously can type 216 pages in 4 hours. In one hour, R can type as many pages more than Q as Q can type more than P. During a period of 5 hours, R can type as many pages as P can during 7 hours. How many pages does each of them type per hour.
- (A) 14, 17, 20 (B) 15, 17, 22 (C) 15, 18, 21 (D) 16, 18, 22
103. The base of a triangle is 15 cm. Two lines are drawn parallel to the base, terminating in the other two sides, and dividing the triangle into three equal areas. The length of the parallel line close to the base is :
- (A) $5\sqrt{6}$ cm (B) $4\sqrt{3}$ cm (C) 10 cm (D) 5.5 cm
104. In triangle ABC the value of $\angle A$ is given by $5\cos A - 3 = 0$, then the equation whose roots are $\sin A$ and $\tan A$ is :
- (A) $15x^2 - 32x - 16 = 0$ (B) $15x^2 + 32x - 16 = 0$
 (C) $15x^2 + 32x + 16 = 0$ (D) $15x^2 - 32x + 16 = 0$
105. The value of $f(x, y) = \left(\frac{\sqrt[4]{x^3y} - \sqrt[4]{xy^3}}{\sqrt{y} - \sqrt{x}} + \frac{1 + \sqrt[4]{xy}}{\sqrt[4]{xy}} \right)^{-2} \cdot \left(1 + 2\sqrt{\frac{y}{x}} + \frac{y}{x} \right)^{\frac{1}{2}}$ when $x = 9, y = 0.04$:
- (A) 0.64 (B) 0.46 (C) 0.32 (D) 0.11
106. Find a single discount equivalent to the discount series of 20%, 10%, 5% ?
- (A) 30% (B) 31.6% (C) 68.4% (D) 35%
107. Find the sum of all three digit numbers that gives a remainder 4 when they are divided by 5.
- (A) 98270 (B) 99270 (C) 102090 (D) 90270
108. The value of $5 \times 2 - [3 - \{5 - (7 + 2 \times 4 - 19)\}]$ is :
- (A) 0 (B) 14 (C) 16 (D) -5

109. Seven equal cubes each of the side 5cm are joined end to end. Find the surface area of the resultant cuboid.
- (A) 750 cm^2 (B) 1500 cm^2 (C) 2250 cm^2 (D) 700 cm^2
110. 8% of the people eligible to vote are between 20 and 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25, actually voted. In that election the number of persons between 20 and 25, who actually voted, was what percentage of those eligible to vote?
- (A) 4.2 (B) 6.4 (C) 6.8 (D) 8
111. A chord AB of a circle of radius 5.25 cm makes an angle 60° at the centre of the circle. Find the area of major segment.
- (A) 168 cm^2 (B) 42 cm^2 (C) 84 cm^2 (D) 78 cm^2
112. A boat covers 32 km upstream and 36 km downstream in 7 hrs. Also, it covers 40 km upstream and 48 km downstream in 9 hrs. Find the speed of the stream.
- (A) 2 km/hr (B) 3 km/hr (C) 1 km/hr (D) 5 km/hr
113. A contractor undertakes to build a wall in 50 days. He employs 50 people for the same. However, after 25 days he finds that the work is only 40% complete. How many more men need to be employed to complete the work 10 days before time.
- (A) 50 men (B) 75 men (C) 100 men (D) 125 men
114. A shopkeeper advertises for selling cloth at 4% loss. However, by using a false meter scale he actually gains 25%. What is the actual length of the scale?
- (A) 76.8 cm (B) 77.8 cm (C) 74.8 cm (D) 75.8 cm

115. A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate it would have fetched Rs.480 more. Find the principal.
- (A) Rs. 4000 (B) Rs. 5000 (C) Rs. 6000 (D) Rs. 3000
116. The number of integral pair (x, y) satisfying the equation : $(x - y)^2 + 2y^2 = 27$ is :
- (A) 8 (B) 7 (C) 6 (D) 5
117. Ram and shyam are two different persons, but when they worked together , they completed a job in 10 days. Had Ram worked at half of his efficiency and shyam at five times of his efficiency, they would have finished the job in 50% of the scheduled time. Shyam can complete the job working alone in y days. Then y is :
- (A) 20 days (B) 30 days (C) 40 days (D) 35 days
118. The area of a rhombus is $\frac{x^2}{2} + 2x + \frac{3}{2}$ ($x \geq 0$) then the smaller diagonal is :
- (A) $x+2$ (B) $x+3$ (C) $x+1$ (D) 1
119. When the price of mobile was reduced by 20% the number of mobiles sold is increased by 40%. The effect on the sale was :
- (A) 12% increase (B) 12% decrease (C) 30% increase (D) 40% decrease
120. A man invested $\frac{1}{3}$ rd of his capital at 7% , $\frac{1}{4}$ th at 8% and remainder at 10%. If his annual income is Rs. 561. Find the capital.
- (A) Rs. 5000 (B) Rs. 6000 (C) Rs. 6600 (D) Rs. 5600

SECTION E : ATTITUDE & LEADERSHIP

121. Needs, setting standards and maintaining discipline, and appointing sub-leaders according to Adair's approach, called as :
- (A) Work functions (B) Task functions
(C) Individual functions (D) Team functions
122. The attitude based on the concept that "every individual will have an Intention to react in a certain way toward something is "under _____ component of Attitude.
- (A) Behavioral (B) Cognitive (C) Affective (D) Positive
123. Transactional leadership has which of the following characteristics ?
- (A) Takes the view that rewards and punishment motivate staff.
(B) Seeks to ensure staffs understands issues facing the organisation
(C) Seeks to involve staff in the decision making process
(D) Believe success arises from leaders and staff working together
124. _____ are the approaches to the study of leadership which emphasise the personality of the leader :
- (A) Contingency theories (B) Group theories
(C) Trait theories (D) Inspirational theories
125. This is a conventional distinction made between managers and leaders.
- (A) Managers maintain while leaders develop.
(B) Managers innovate while leaders administer.
(C) Managers inspire while leaders control.
(D) Managers originate while leaders imitate.
126. _____ used the terms "employee-centred" and "production-centred" to describe leader behaviour.
- (A) Blake and McCanse (B) Fiedler
(C) McGregor (D) Likert

127. Which of the following theories posits that, under uncertainty, people infer their attitudes from their own behavior ?
- (A) Implicit theory (B) Self-perception theory
(C) Temporal self-appraisal theory (D) Self-categorization theory
128. What does it mean that an attitude is a hypothetical mediating variable ?
- (A) It can be observed directly (B) It cannot be observed directly
(C) It links a stimulus to a response (D) Both (B) and (C)
129. Attitude is defined as "a learned predisposition to respond in a consistently _____ with respect to a given object."
- (A) Favourable or unfavourable manner
(B) Favourable manner
(C) Unfavourable manner
(D) Favourable and unfavourable manner
130. A pessimist sees the difficulty in every opportunity; an optimist sees the opportunity in every difficulty. " These words were said by :
- (A) Winston Churchill (B) A. P. J. Abdul Kalam
(C) S. Robbins (D) Dressler
131. Which is **NOT** true of charismatic leaders ?
- (A) Charismatic leaders have a vision.
(B) Charismatic leaders are focused on their personal needs.
(C) Charismatic leaders are willing to take high personal risk.
(D) Charismatic leaders have behavior that is unconventional.
132. You are asked by your instructor to discuss about Contingency theory. What should you be included in your discussion ?
- (A) It is a theory in leadership that views the pattern of leader behaviour as dependent on the interaction of the personality and the needs of the situation.
(B) It emphasizes that both leaders and followers should act on one another to raise their motivation.
(C) It states that leadership qualities inspire followers to be motivated by what they do.
(D) None of the options

133. LBDQ stands for :
- (A) Leader Behavior Description Questionnaire
 - (B) Leader Behavior Defining Questionnaire
 - (C) Leader Behavior Dominance Questionnaire
 - (D) None of these
134. Which of the following is not a result of Positive attitude of employee ?
- (A) Improved productivity
 - (B) Successful Organisation
 - (C) Higher performance
 - (D) None of the options
135. The Ohio State Leadership Studies revealed _____ and initiating structure as two major dimensions of leadership behaviour :
- (A) Control
 - (B) Communication
 - (C) Collaboration
 - (D) Consideration
136. _____ is not a capability of an employee having Positive attitude
- (A) Focus
 - (B) Creativity
 - (C) Pessimism
 - (D) Confidence
137. A Charismatic leader's _____ is the key to follower acceptance.
- (A) Energy
 - (B) History with the organization
 - (C) Credentials
 - (D) Vision
138. Attitude conviction involves which of the following elements ?
- (A) Ego preoccupation
 - (B) Centrality
 - (C) Importance
 - (D) Strength
139. Leadership is often associated more with _____.
- (A) Consistency
 - (B) Planning
 - (C) Paperwork
 - (D) Change
140. Participative leadership has which of the following characteristics ?
- (A) Believe success arises from leaders and staff working together
 - (B) Employs a clear chain of command
 - (C) Takes the view that rewards and punishment motivate staff
 - (D) Seeks to involve staff in the decision making process

141. Who is a visionary leader ?
- (A) One who has medium term perspective
 - (B) One who has short term perspective
 - (C) One who has long term perspective
 - (D) One who takes initiative for need satisfaction
142. This is a joint trait of a leader which is defined as his ability to possess honesty, responsibility and maturity in the working area.
- (A) Integrity
 - (B) Personality
 - (C) Intelligence
 - (D) Flexibility
143. Person perception is :
- (A) The process of how we manage, experience, and adjust our emotions
 - (B) The process by which we form impressions and process information about people
 - (C) The process by which we adjust to different cultural situations
 - (D) The process by which we learn cultural norms and standards of behaviour
144. Which style of leadership focuses on goals, standards, and organization ?
- (A) Task leadership
 - (B) Social leadership
 - (C) Semantic leadership
 - (D) Transformational leadership
145. Which of the following terms are used to describe the fact that attitudes must be inferred from an individual's overt response to a stimulus ?
- (A) Cognitive consistency
 - (B) Hypothetical mediating variable
 - (C) Selective perception
 - (D) Selective interpretation
146. The two types of attitude in a work place can be termed as_____.
- (A) Favourable and unfavourable
 - (B) Optimistic and Pessimistic
 - (C) Individual and group
 - (D) Satisfied and dissatisfied

147. A satisfied employee will be a :

- (A) Motivator to others (B) Manager
(C) High performer (D) Team Leader

148. Which can be considered as a difficulty in changing the Attitude of the employees ?

- (A) Escalation of Commitment (B) Providing new information
(C) Use of fear (D) Influence by friends and peers

149. What make you think is the most effective leadership style that can be used during emergency situations ?

- (A) Democratic (B) Laissez-fire (C) Autocratic (D) Supportive

150. Belief, opinion, knowledge, emotions feelings intentions are the components of _____.

- (A) OB (B) Job satisfaction (C) Attitude (D) Personality

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